

$$T(n) = T\left(\frac{n}{3}\right) + \underline{5n}$$

ESPERTO

$$f(n) = 5n$$

$$n^{\log_a b} = n^{\log_3 1} = n^0 = 1$$

$$f(n) = \Omega\left(n^{\log_3 1 + \epsilon}\right)$$

CASO 3

$$5n^1 = \Omega(n^\epsilon) \quad \epsilon = 0.5$$

$$\Rightarrow \cancel{\frac{5n}{3}} \leq \cancel{5c} n$$

$$\frac{1}{3} \leq c$$

$$\Theta(f(n)) = \Theta(n)$$

$$T(n) = T\left(\frac{n}{2}\right) + \log n \quad \leftarrow$$

IPOTESI: $O((\log n)^2)$? VERO

$$\underline{T\left(\frac{n}{2}\right) = O\left(\left(\log\left(\frac{n}{2}\right)\right)^2\right)} \quad \leftarrow \text{VERO}$$

$$\underbrace{O(f(n))}_{\uparrow} \leq \underline{C \cdot f(n)}$$

$$T(n) = T\left(\frac{n}{2}\right) + \log n$$

$$\leq C \cdot \left(\log \frac{n}{2}\right)^2 + \log n$$

$$C \cdot (\log_2 n - \log_2 2)^2 + \log_2 n$$

$$C \cdot (\log n - 1)^2 + \log n$$

$$T(n) \leq \underline{C \cdot (\log n)^2 + C - 2C \log n + \log n}$$

$$T(n) \leq C \cdot (\log n)^2$$

DA VERIFICARE



$$T(n) \leq \cancel{c(\log n)^2} + c - \underbrace{2c \log n + \log n}_{\log n(2c-1)} \leq \cancel{c(\log n)^2}$$

$$- \underbrace{\log n}_{\log n} (2c-1) + c \leq 0$$

$$\Rightarrow (2c-1) \cdot \underbrace{\log n}_{\log n} - c \geq 0$$

$$\underbrace{n > 2} \Rightarrow \underbrace{\log n}_{\log n} > \underbrace{1}_{1}$$

$$(2c-1) \cdot \underbrace{1}_{1} - c \geq 0$$

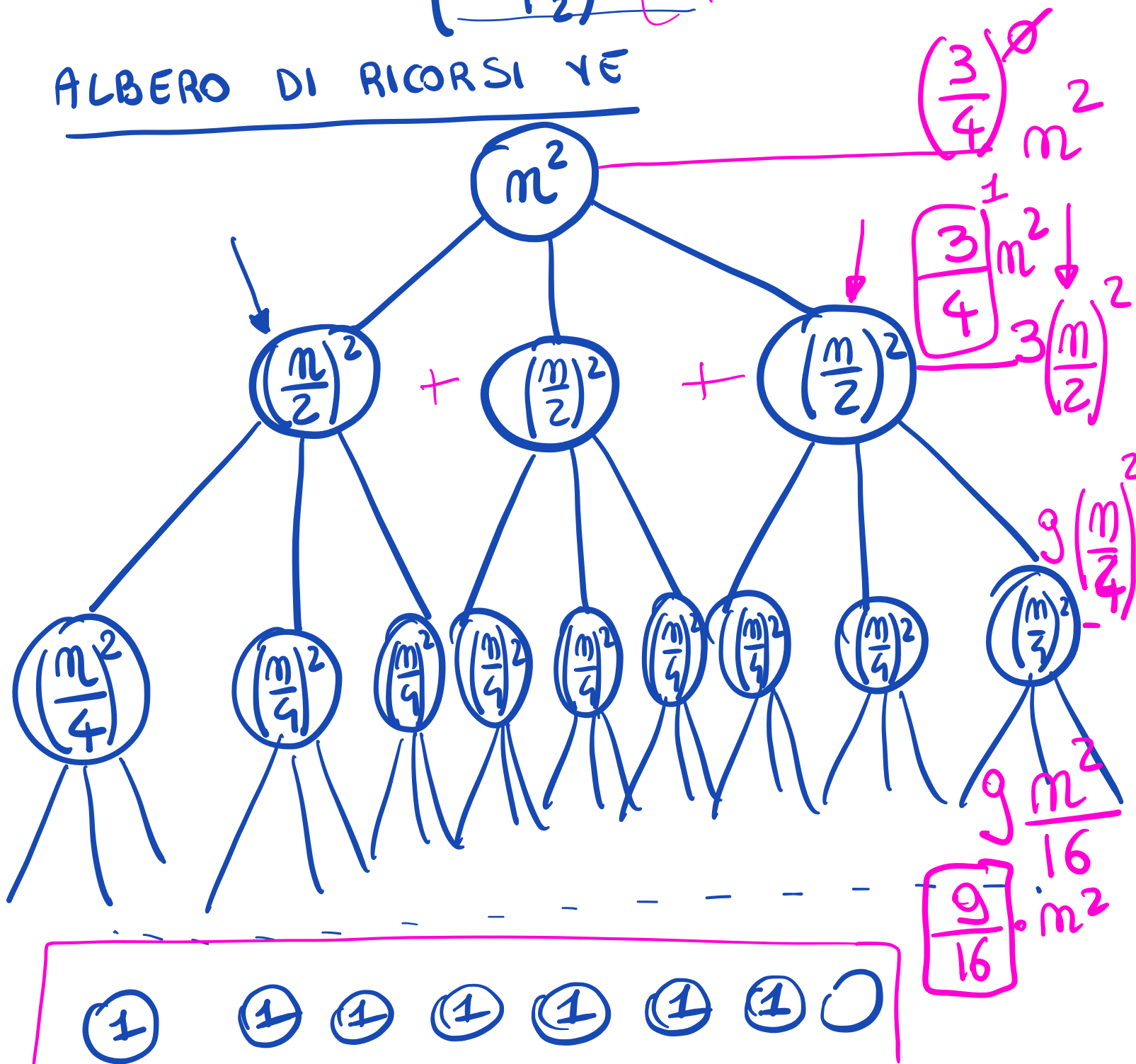
$$\cancel{2c} - 1 - \cancel{c} \geq 0$$

$$c \geq 1$$

$$\boxed{c=2}$$

$$T(n) = \begin{cases} 1 & \text{se } n=1 \\ 3T\left(\frac{n}{2}\right) + n^2 & \text{se } n > 1 \end{cases}$$

ALBERO DI RICORSI



$$\frac{n}{2^i} = 1 \Rightarrow n = 2^i$$

$$\log n = \log 2^i$$

$$\Rightarrow \boxed{i = \log m}$$

$$\sum_{i=0}^{\log m - 1} \left(\frac{3}{4} \right)^i \cdot m^2$$

$$m^2 \cdot \sum_{i=0}^{\log m - 1} \left(\frac{3}{4} \right)^i \quad \leftarrow$$

$$m^2 \cdot \left(\frac{1 - \left(\frac{3}{4} \right)^{\log m}}{1 - \left(\frac{3}{4} \right)} \right) \Rightarrow$$

$$4m^2 \cdot \left(1 - \left(\frac{3}{4} \right)^{\log m} \right)$$

$\bigcirc (m^2)$

